

Team Report

Self-Healing O-RAN Timing Security

Fronthaul S-plane · final software checkpoint · plain language, no background needed

1 The problem

A 5G radio network needs its computers and radios to agree on time within roughly **100 nanoseconds**. Faulty or malicious timing can disrupt service in about **two seconds**. Most security tools stop at raising an alarm. This project goes further: it estimates whether an event is a normal fault, a known attack, or something unfamiliar; checks recovery options in a digital twin; and commits a safe action before the deadline.

2 What we built

- A repeatable simulator for PTP, SyncE, GNSS, clock holdover, network delay and oscillator behaviour.
- Benign faults and attacks, each with a realistic benign *look-alike* so the tests are genuinely hard.
- A **28-feature** detector: timing, network behaviour, PTP grandmaster changes, receiver status, oscillator consistency.
- A novelty layer that routes unfamiliar events to a safe default (labelled UNKNOWN).
- A **2-of-3 voting rule** so a single noisy 0.2 s window can't trigger an action.
- A digital twin that compares recovery choices before the system acts.
- Real packet-capture readers, public TIMESAFE evaluation, and a Linux PTP/netem test harness.

The automated suite contains **39 passing tests** and rebuilds every result with one command.

3 Final measured results

“Protection” means the attack was either recognised or safely treated as unfamiliar.

Test	Protection	Detected within 2 s
Simulated spooof	93.30%	100%
Simulated replay	80.45%	100%
Simulated message-flood DoS	88.27%	100%
Simulated GNSS jamming	91.57%	100%
Strict unseen healthy-looking GNSS spooof	0.00%	0%
Real TIMESAFE Announce attack	99.96%	100%
Real TIMESAFE Follow-Up attack	99.66%	100%
Real TIMESAFE Single-Step attack	99.66%	100%

The 2-of-3 rule cut real benign false alarms from **4.49% to 2.37%** without costing the two-second deadline.

4 What worked

- Watching how PTP **grandmaster attributes change** raised real Announce-attack protection from 23.9% to ~100%.
- **Separate novelty detectors** per signal group recovered attacks the ordinary classifier missed — notably message flooding and replay.

- The **2-of-3 rule** roughly halved noisy alarms at no cost to the deadline.
- Comparing reported GNSS state against **physical oscillator behaviour** solved most unseen GNSS jamming.

5 What did not work (reported honestly)

- A GNSS receiver saying “I am synchronised” **cannot be trusted on its own** — a spoofed receiver still reports healthy.
- A spoof that behaves exactly like healthy timing stayed **invisible** when its family was strictly excluded from training.
- Comparing GNSS, network PTP and a peer source added **no** protection and hurt some established results. Those seven features are kept for research but **switched off by default**.
- If every time source is compromised the same way, they still agree with each other — relative agreement adds **0.00 percentage points**.

Note: the multi-source experiment kept related GNSS attacks in training, so its high single-source figure does *not* overturn the strict 0% unseen-spoof result.

6 What this means

Software can detect protocol manipulation, unfamiliar traffic, timing jams and many physics inconsistencies. It **cannot prove that a single healthy-looking clock is telling the truth** when no independent trusted reference exists. Solving that requires something the attacker cannot forge — authenticated GNSS (e.g. Galileo OSNMA), a physically independent clock, or hardware-backed comparison between genuinely independent references.

7 Honest limitations

- No hardware timestamp card, real clock, authenticated GNSS receiver, O-DU or O-RU has been tested.
- Public captures contain few independent attacks and **no benign planned grandmaster change**, so deployment false alarms may be higher.
- SyncE quality still needs a live syncE4L or radio-unit telemetry feed.
- A model trained on one real attack style can fail completely on another.
- These are research measurements, **not telecom certification**.

8 Decision and next step

Decision & next step: software feature work is **closed** — further single-source features now give diminishing returns. The next phase is **Tier 3 hardware validation**: authenticated GNSS, physically independent clocks, hardware timestamps, longer normal-operation recordings and controlled authorised attacks. That phase also carries the clearest application path: a vendor-neutral timing-security validation product for telecom labs, private 5G operators and equipment vendors.